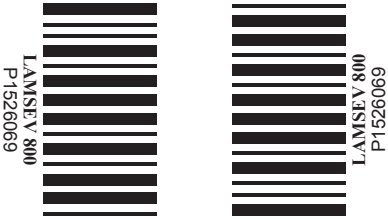


Pharmacode position may change as per Supplier’s m/c requirement & additional small pharma code may appear on the front / back panel



1. NAME OF THE DRUG PRODUCT

LAMSEV 800 (Sevelamer Carbonate Tablets 800 mg)

PHARMACEUTICAL DOSAGE FORM

Oral Film Coated Tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITIONS

Sevelamer Carbonate Tablets 800 mg

Each film-coated tablet contains 800 mg of Sevelamer Carbonate.

For the full list of excipients, see section 6.1

3. PHARMACEUTICAL FORM

Product Description

White to off white, oval shaped, film-coated tablets, imprinted with “J 75” on one side with edible black ink and plain on other side.

4. CLINICAL PARTICULARS

4.1 THERAPEUTIC INDICATIONS

LAMSEV 800 is indicated for the control of hyperphosphatemia in adult patients receiving haemodialysis or peritoneal dialysis.

LAMSEV 800 is also indicated for the control of hyperphosphatemia in adult patients with chronic kidney disease not on dialysis with serum phosphorus ≥ 1.78 mmol/l.

LAMSEV 800 should be used within the context of a multiple therapeutic approach, which could include calcium supplement, 1,25-dihydroxy Vitamin D3 or one of its analogues to control the development of renal bone disease.

4.2 POSOLOGY AND METHOD OF ADMINISTRATION

Because of the rapid disintegration of the carbonate salt tablet and its rapid reaction with the hydrochloric acid in the stomach, the dosing of LAMSEV 800 is anticipated to be similar to that of the hydrochloride salt.

LAMSEV 800 should be given three times a day with meals.

Patients Not Taking a Phosphate Binder: The recommended starting dose of LAMSEV 800 is 800 to 1600 mg, which can be administered as one or two LAMSEV 800 mg Tablets, with meals based on serum phosphorus level. Table 1 provides recommended starting doses of LAMSEV 800 for patients not taking a phosphate binder.

Table 1. Starting Dose for Patients Not Taking a Phosphate Binder

Serum Phosphorus	LAMSEV 800
> 1.78 and < 2.42 mmol/L	1 tablet three times daily with meals
≥ 2.42 and < 2.91 mmol/L	2 tablets three times daily with meals
≥ 2.91 mmol/L	2 tablets three times daily with meals

Patients Switching From Sevelamer Hydrochloride: For patients switching from sevelamer hydrochloride, sevelamer carbonate should be prescribed on a gram per gram basis. Further titration to the desired phosphate levels may be necessary. The highest daily dose of sevelamer carbonate studied was 14 grams in CKD patients on dialysis.

Patients Switching From Calcium Acetate: In a study in 84 CKD patients on hemodialysis, a similar reduction in serum phosphorus was seen with equivalent doses (approximately mg for mg) of sevelamer hydrochloride and calcium acetate. Table 2 gives recommended starting doses of LAMSEV 800 based on a patient’s current calcium acetate dose.

Table 2. Starting Dose for Dialysis Patients Switching From Calcium Acetate to LAMSEV 800

Calcium Acetate 667 mg (Tablets per meal)	LAMSEV 800 (Tablets per meal)
1 tablet	1 tablet
2 tablets	2 tablets
3 tablets	3 tablets

Dose Titration for All Patients Taking LAMSEV 800. The dose should be increased or decreased by one tablet per meal at two-week intervals, as necessary, with the goal of controlling serum phosphorus within the target range of 1.13 mmol/L to 1.78 mmol/L.

INSTRUCTIONS FOR USE

For oral use. Tablets should be swallowed intact and should not be crushed, chewed, or broken into pieces prior to administration. LAMSEV 800 should be taken with food and not on an empty stomach.

4.3 CONTRAINDICATIONS

LAMSEV 800 is contraindicated in patients with hypophosphatemia or bowel obstruction.

LAMSEV 800 is contraindicated in patients with hypersensitivity to sevelamer carbonate or to any of the excipients.

4.4 SPECIAL WARNINGS AND PRECAUTIONS FOR USE

The safety and efficacy of sevelamer carbonate have not been established in adult patients with chronic kidney disease not on dialysis with serum phosphorus < 1.78 mmol/l. Therefore, it is currently not recommended for use in these patients.

The safety and efficacy of sevelamer carbonate have not been established in patients with the following disorders:

- Dysphagia
- Swallowing disorders
- Severe gastrointestinal motility disorders including untreated or severe gastroparesis, retention of gastric contents and abnormal or irregular bowel motion
- Active inflammatory bowel disease
- Major gastrointestinal tract surgery

Treatment of these patients with LAMSEV 800 should only be initiated after careful benefit/risk assessment. If the therapy is initiated, patients suffering from these disorders should be monitored. LAMSEV 800 treatment should be re-evaluated inpatients who develop severe constipation or other severe gastrointestinal symptoms.

Intestinal obstruction and ileus/subileus

In very rare cases, intestinal obstruction and ileus/subileus have been observed in patients during treatment with sevelamer hydrochloride (capsules/tablets), which contains the same active moiety as sevelamer carbonate. Constipation may be a preceding symptom. Patients who are constipated should be monitored carefully while being treated with LAMSEV 800. The treatment should be re-evaluated in patients who develop severe constipation or other severe gastrointestinal symptoms.

Fat soluble vitamins and folate deficiency

Patients with CKD may develop low levels of fat-soluble vitamins A, D, E and K, depending on dietary intake and the severity of their disease. It cannot be excluded that sevelamer carbonate can bind fat-soluble vitamins contained in ingested food. In patients not taking supplemental vitamins but on sevelamer, serum vitamin A, D, E and K status should be assessed regularly. It is recommended that vitamin supplements be given if necessary. It is recommended that CKD patients not on dialysis are given vitamin D supplements (approximately 400 IU of native vitamin D daily) which can be part of a multivitamin preparation to be taken apart from their dose of sevelamer carbonate. In patients undergoing peritoneal dialysis additional monitoring of fat-soluble vitamins and folic acid is recommended, since vitamin A, D, E and K levels were not measured in a clinical study in these patients.

There is at present insufficient data to exclude the possibility of folate deficiency during long term sevelamer carbonate treatment. In patients not taking supplemental folic acid but on sevelamer, folate level should be assessed regularly.

Hypocalcaemia/hypercalcaemia

Patients with CKD may develop hypocalcaemia or hypercalcaemia. Sevelamer carbonate does not contain any calcium. Serum calcium levels should therefore be monitored at regular intervals and elemental calcium should be given as a supplement if required.

Metabolic acidosis

Patients with CKD are predisposed to developing metabolic acidosis. As part of good clinical practice, monitoring of serum bicarbonate levels is therefore recommended.

Peritonitis

Patients receiving dialysis are subject to certain risks for infection specific to dialysis modality. Peritonitis is a known complication in patients receiving peritoneal dialysis and in a clinical trial with sevelamer hydrochloride, a greater number of peritonitis cases were reported in the sevelamer group than in the control group. Patients on peritoneal dialysis should be closely monitored to ensure the correct use of appropriate aseptic technique with the prompt recognition and management of any signs and symptoms associated with peritonitis.

Swallowing and choking difficulties

Many of these cases involved patients with co-morbid conditions including swallowing disorders or oesophageal abnormalities. Proper swallowing ability should be carefully monitored in patients with co-morbid conditions. The use of sevelamer carbonate powder in patients with a history of difficulty swallowing should be considered.

Hypothyroidism

Closer monitoring of patients with hypothyroidism co-administered with sevelamer carbonate and levothyroxine is recommended.

Hyperparathyroidism

Sevelamer carbonate is not indicated for the control of hyperparathyroidism. In patients with secondary hyperparathyroidism sevelamer carbonate should be used within the context of a multiple therapeutic approach, which could include calcium as supplements, 1,25-dihydroxy Vitamin D₃ or one of its analogues to lower the intact parathyroidhormone (iPTH) levels.

Inflammatory gastrointestinal disorders

Cases of serious inflammatory disorders of different parts of the gastrointestinal tract (including serious complications such as haemorrhage, perforation, ulceration, necrosis, colitis and colonic/ caecal mass) associated with the presence of sevelamer crystals have been reported. Inflammatory disorders may resolve upon sevelamer discontinuation. Sevelamer carbonate treatment should be re-evaluated in patients who develop severe gastrointestinal symptoms.

4.5 Interaction with other medicinal products and other forms of interaction

Dialysis

Interaction studies have not been conducted in patients on dialysis.

Ciprofloxacin

Decreased bioavailability of ciprofloxacin by approximately 50% when co-administered with sevelamer hydrochloride in a single dose. Consequently, sevelamer carbonate should not be taken simultaneously with ciprofloxacin.

Ciclosporin, mycophenolate mofetil and tacrolimus in transplant patients

Reduced levels of ciclosporin, mycophenolate mofetil and tacrolimus have been reported in transplant patients when co-administered with sevelamer hydrochloride without any clinical consequences (e.g., graft rejection). The possibility of an interaction cannot be excluded and a close monitoring of blood concentrations of ciclosporin, mycophenolate mofetil and tacrolimus should be considered during the use of combination and after its withdrawal.

Levothyroxine

Very rare cases of hypothyroidism have been reported in patients co-administered with sevelamer hydrochloride, which contains the same active moiety as sevelamer carbonate, and levothyroxine. Closer monitoring of thyroid stimulating hormone (TSH) levels is therefore recommended in patients receiving sevelamer carbonate and levothyroxine.

A/s: 210 x 300 mm ■ Black

 AUROBINDO Packaging Development		Product Name	Component	Item Code	Date & Time
		LAMSAV	Leaflet_SPC	P1526069	19.05.2022 & 04.00 pm
		Country	Version No.	Reason of Issue	Reviewed / Approved by
		Malayisa_Unit 7	04	Submission	
Team Leader	Kiran Kumar	Dimensions (mm)	Colours: 01		
Initiator	Shirisha	A/s : 210 x 300 mm	Font Type : Times New Roman Font Size : 7 PT		
Artist	Sree Designers	Pharma Code: 26069			
Additional Information: Word Replacement : 03.05.2022			 26069		

Anti-arrhythmics and anti-seizure medicinal products

Patients taking anti-arrhythmic medicinal products for the control of arrhythmias and anti-seizure medicinal products for the control of seizure disorders were not studied. Therefore, possible reduction in absorption cannot be excluded. The anti-arrhythmic medical product should be taken at least one hour before or three hours after LAMSEV 800, and blood monitoring can be considered.

Proton pump inhibitors

During post-marketing experience, very rare cases of increased phosphate levels have been reported in patients taking proton pump inhibitors co-administered with sevelamer carbonate. Caution should be exercised when prescribing PPI to patients concomitantly treated with LAMSEV 800. The phosphate serum level should be monitored and the LAMSEV 800 dosage adjusted consequently.

Bioavailability

Sevelamer carbonate is not absorbed and may affect the bioavailability of other medicinal products. When administering any medicinal product where a reduction in the bioavailability could have a clinically significant effect on safety or efficacy, the medicinal product should be administered at least one hour before or three hours after sevelamer carbonate, or the physician should consider monitoring blood levels.

Digoxin, warfarin, enalapril or metoprolol

No effect on the bioavailability of digoxin, warfarin, enalapril or metoprolol.

4.6 FERTILITY, PREGNANCY AND LACTATION

Pregnancy

There are no or limited amount of data from the use of sevelamer in pregnant women. Sevelamer has also been shown to reduce the absorption of several vitamins including folic acid. The potential risk to humans is unknown. Sevelamer carbonate should only be given to pregnant women if clearly needed and after a careful risk/benefit analysis has been conducted for both the mother and the foetus.

Breast-feeding

It is unknown whether sevelamer/metabolites are excreted in human milk. The non-absorbed nature of sevelamer indicates that excretion of sevelamer in breast milk is unlikely. A decision on whether to continue/discontinue breast-feeding or to continue/discontinue therapy with sevelamer carbonate should be made taking into account the benefit of breast-feeding to the child and the benefit of sevelamer carbonate therapy to the woman.

Fertility

There is no data from the effect of sevelamer on fertility in humans.

4.7 EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Sevelamer has no or negligible influence on the ability to drive and use machines.

4.8 UNDESIRABLE EFFECTS

Summary of the safety profile

The most frequently occurring adverse reactions were all in the gastrointestinal disorders system organ class. Most of these adverse reactions were mild to moderate in intensity.

MedDRA System Organ Class	Very Common	Common	Very Rare	Not known
Immune System Disorders			Hypersensitivity	
Gastrointestinal disorders	Nausea, vomiting, upper abdominal pain, constipation	Diarrhoea, dyspepsia, flatulence, abdominal pain		Intestinal obstruction, ileus/subileus, intestinal perforation, gastrointestinal haemorrhage, intestinal ulceration, gastrointestinal necrosis, colitis, intestinal mass
Skin and subcutaneous tissue disorders				Pruritus, rash
Investigations				Crystal deposit intestine

Paediatric population

In general, the safety profile for children and adolescents (6 to 18 years of age) is similar to the safety profile for adults.

4.9 OVERDOSE

Sevelamer hydrochloride, which contains the same active moiety as sevelamer carbonate, has been given in doses of up to 14 grams per day for eight days with no adverse reactions. In CKD patients, the maximum average daily dose studied was 14.4 grams of sevelamer carbonate in a single daily dose.

The symptoms observed in case of overdose are similar to adverse reactions listed in section 4.8, including mainly constipation and other known gastrointestinal disorders.

Appropriate symptomatic treatment should be provided.

5. PHARMACOLOGICAL PROPERTIES

5.1 PHARMACODYNAMIC PROPERTIES

Pharmacotherapeutic group: All other therapeutic products, drugs for treatment of hyperkalaemia and hyperphosphatemia.

ATC Code: V03AE02

Mechanism of action

LAMSEV 800 contains sevelamer, a non-absorbed phosphate binding crosslinked polymer, free of metal and calcium. Sevelamer contains multiple amines separated by one carbon from the polymer backbone which become protonated in the stomach. These protonated amines bind negatively charged ions such as dietary phosphate in the intestine.

Pharmacodynamic effect

By binding phosphate in the gastrointestinal tract and decreasing absorption, sevelamer lowers the phosphorus concentration in the serum. Regular monitoring of serum phosphorus levels is always necessary during phosphate binder administration.

Efficacy and safety

Sevelamer carbonate in both tablet and powder formulations when administered three times per day has been shown to be therapeutically equivalent to sevelamer hydrochloride and therefore effective in controlling serum phosphorus in CKD patients on haemodialysis.

Sevelamer has been shown to bind bile acids. Bile acid binding by ion exchange resins is a well-established method of lowering blood cholesterol.

Because sevelamer binds bile acids, it may interfere with the absorption of fat soluble vitamins such as A, D, E and K.

Sevelamer does not contain calcium and decreases the incidence of hypercalcaemic episodes as compared to patients using calcium based phosphate binders alone. The effects of sevelamer on phosphorus and calcium were proven to be maintained throughout a study with one year follow-up. This information was obtained from studies in which sevelamer hydrochloride was used.

Paediatric population

Most of adverse events reported as related, or possibly related, to sevelamer carbonate were gastrointestinal in nature. No new risks or safety signals were identified with the use of sevelamer carbonate.

5.2 PHARMACOKINETIC PROPERTIES

Pharmacokinetic studies have not been carried out with sevelamer carbonate. Sevelamer hydrochloride, which contains the same active moiety as sevelamer carbonate, is not absorbed from the gastrointestinal tract.

No evidence of accumulation of sevelamer was seen. However, the potential absorption and accumulation of sevelamer during long-term chronic treatment (> one year) cannot be totally excluded.

6. PHARMACEUTICAL PARTICULARS

6.1 LIST OF EXCIPIENTS

Mannitol, Crospovidone, Hydroxypropyl Cellulose, Colloidal Silicon Dioxide, Talc, Glyceryl Behenate, Aquarius Prime BBN119000 Clear and Opacode Black S-1-17823.

6.2 STORAGE CONDITION

Store below 30°C. Protect from moisture.

6.3 SHELF LIFE

24 months.

6.4 NATURE AND CONTENTS OF CONTAINER

White opaque HDPE Bottle with a polypropylene cap containing 30 tablets (with outer carton) and 180 tablets (without the outer carton).

All pack sizes may not be marketed.

7. MANUFACTURER & PRODUCT REGISTRATION HOLDER

Manufacturer

Aurobindo Pharma Ltd.,
Unit VII, SEZ, TSIC, Plot No., S1,
Survey No's: 411/P, 425/P, 434/P, 435/P & 458/P,
Green Industrial Park, Polepally Village,
Jedcherla Mandal, Mahaboobnagar District,
Telangana State, 509302, India.

Product Registration Holder

Synerrv Sdn. Bhd.,
SO-29-2, Menara 1,
KL Eco City, Jalan Bangsar,
Kg. Haji Abdullah Hukum,
59200 Kuala Lumpur, W.P.
Kuala Lumpur, Malaysia.

8. DATE OF PACKAGE INSERT REVISION

May 2022